

7 Animation

So far, we've been working in the three dimensions of space. Time adds a fourth dimension to your work. Changing an object over time is called animation, and animation is what truly brings your scene to life. The way an object moves tells the audience a lot about its size and weight and, for characters, its personality. 3ds Max has a wealth of tools for creating and sculpting motion. Learning how to use the tools, though, is only the first step in the process. An animator must understand how the laws of motion work, as well as how the audience perceives motion. Character animators also have to learn how motion affects a character's personality and mood.

7.1 Creating animation

3ds Max offers a variety of methods for creating animation. Each method has its advantages. Understanding each method will let you decide which approach to take when animating a scene. Creating keys is the most common way of animating within 3ds Max. A key simply records a parameter's value at a given point in time. By changing the value over time, you create motion and animation. Keys allow you a wide degree of control over every parameter of an object, from its position to its colour, and just about everything else. To edit keys, you use the Curve Editor and the Dope Sheet. Both allow you to control timing precisely.

Attributes such as position, rotation, and scale can be animated to control how an object moves through space. Any attribute (such as this object's colour) attached to an object can be animated. Parametric animation uses a set of user-defined parameters to create motion. Constraints can match the motion of one object to another, such as this car to the path defined by a curve.

7.2 Using controllers

Controllers are the heart of animation in 3ds Max. Everything that can be animated in 3ds Max is handled by a controller. A controller is a motion plug-in that handles the motion data created through animation. Constraints, as described in the previous chapter, are also driven by controllers. Some controllers can be key framed, others use scripts or mathematical formulas, and others create motion based on user-defined parameters. Much like modifiers used in modelling, controllers can be stacked in a list so that